Rahul Shelke

Mumbai, Maharashtra — Portfolio: rahul-shelke-1.github.io
Email: rahulshelke3399@gmail.com — Phone: +91-9130619363
LinkedIn: linkedin.com/in/rahulshelke19 — GitHub: github.com/rahul-shelke-1

Professional Summary

Machine Learning Engineer with industry experience in NLP product development and end-to-end ML systems. Skilled in Python, model development, cloud deployment, and MLOps, with a focus on building scalable, production-oriented AI/ML applications.

Technical Skills

Programming: Python, SQL

Machine Learning: Statistics, Scikit-learn, TensorFlow, Model Evaluation, Feature Engineering, NLP, Generative AI, LLMs

Data Engineering & Analytics: Pandas, NumPy, MySQL, MongoDB, Apache Airflow, Tableau, Matplotlib, Seaborn

Backend & GenAI: FastAPI, Flask, Streamlit, Hugging Face, LangChain

Cloud, IaC & MLOps: AWS, Docker, GitHub, OpenTofu, MLflow

Professional Experience

Independent Data Science Practitioner — *Mumbai, Maharashtra*
Independent Projects — *March 2025 – Present*

- Developing end-to-end data and ML systems spanning data ingestion, transformation, feature engineering, model development, and inference applications.
- Building engineering workflows using **Python**, **Docker**, **FastAPI**, and **AWS**, incorporating infrastructure as code, CI/CD, model lifecycle management, and observability.

Project-Based Intern — *Remote*
iNeuron.ai — *May 2024 – Dec 2024*

- Developed two data science applications covering data preprocessing, model development, and cloud deployment.

Associate Data Scientist — *Bengaluru, Karnataka*
Altimetrik — *June 2022 – Jan 2023*

- Developed intent classification capabilities using **RASA** for an NLP-based conversational analytics product.
- Built Transformer-based classifiers for emotion, sarcasm, and urgency detection, achieving **F1 scores above 90%**; trained models on GPU-enabled **AWS EC2** instances and used **Docker** for containerization.
- Applied **LDA** topic modeling and time-series forecasting to develop Twitter trend-analysis capabilities for an NLP-based product.

Project Intern — *Bengaluru, Karnataka*
Altimetrik — *Jan 2022 – June 2022*

- Translated operational challenges into data science problem statements, developed **Flask/Plotly** dashboards, and contributed to an NLP intent-classification proof of concept.

Data Science Intern — *Mumbai, Maharashtra*
Railofy – Rodeo Travel Technologies — *Aug 2021 – Nov 2021*

- Built model-monitoring and selection dashboards using **Kibana**, **Python**, and **Streamlit**; tracked model performance, data drift, and target drift using **Elasticsearch** and **MySQL**.

Selected Projects

Heart Stroke Risk Stratification Platform — *GitHub*

- Designed an end-to-end ML platform for stroke risk prediction and risk-based resource allocation, integrating data ingestion, preprocessing, feature engineering, and model development.
- Developed cloud-oriented ML workflows using **AWS SageMaker**, **Docker**, and **Terraform/OpenTofu**, with model lifecycle management and monitoring infrastructure.

BBC News Data Engineering Pipeline — *GitHub*

- Built a news ingestion and ETL pipeline using **Python**, **Selenium**, and **BeautifulSoup**, storing raw content in **MongoDB** and transforming structured data into **PostgreSQL**.
- Integrated **RabbitMQ** for message processing and implemented observability using **Prometheus**, **Grafana**, and **Loki**.

Education & Certifications

Full Stack Data Science Masters — *iNeuron.ai* — *Apr 2024 – Dec 2024*
University of Mumbai — *B.E. in Computer Engineering (CGPA: 7.32)* — *Aug 2016 – Nov 2020*